ROADWATCH

Project Presentation Notes and Validation Checklist

Design of Smart Cities | Tamil Nadu Smart Cities Project | Sidak Singh | VIT Chennai

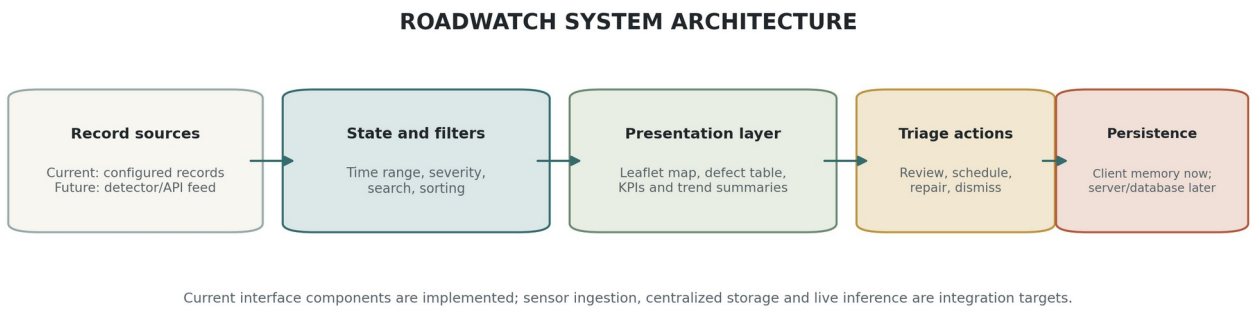
1. Opening (30-45 seconds)

“Roadwatch is a geospatial dashboard for organizing road-defect records for municipal review. It brings location markers, severity, timestamps, work status, workload indicators, and record-level actions into one interface. The current focus is the operator workflow: inspect a location, find the relevant record, and track its maintenance status.”

2. Problem and objective (45 seconds)

Road defects need to be reviewed by location and urgency. A map alone does not manage follow-up, while a table alone hides spatial patterns. Roadwatch links both views and adds severity summaries and maintenance-state actions. The intended use is to help municipal teams prioritize review and coordinate maintenance.

3. Architecture and workflow (60 seconds)



Current interface components are implemented; sensor ingestion, centralized storage and live inference are integration targets.

Explain the sequence: records -> shared filters -> map/table and summaries -> detail drawer -> status action. Be explicit that the current data records are configured in the page and status updates are held in browser memory. A real deployment needs an authenticated API, database, and verified upstream data.

4. Live interface walkthrough (90 seconds)

- Open the overview and point out the KPI cards and critical-attention banner.
- Use the 24-hour, 7-day and 30-day controls and explain how the current time window scopes the records.
- Search by a road name and then by a defect identifier.
- Use severity/status chips and sort by confidence or severity.
- Click a map marker and then a table row; point out that both open the same record drawer.
- For a record, show scheduling, repair, dismissal or reopening actions. Explain that the current browser version does not persist updates to a server.
- Point to the map attribution and explain that road positions must come from validated field coordinates before operational dispatch.

5. Implemented vs. required next steps

Implemented in current source	Required before operational rollout
Leaflet basemap integration and severity-coded markers	Verified GPS feed and coordinate-quality checks

Implemented in current source	Required before operational rollout
Linked map, table, search, filters and sorting	Backend API, persistent database and audit history
KPI, severity mix, trend display and record drawer	Authenticated roles and permission checks
Client-side maintenance status actions	Crew dispatch integration and notifications
Configured defect record fields	Validated upstream detector/reporting pipeline and recorded accuracy tests

6. Questions likely to be asked

What is the main contribution? A linked operator interface that unifies geospatial inspection and maintenance-status triage.

Does Roadwatch detect potholes using AI? The current source does not embed a trained vision model. It presents structured road-defect records; a detector can be integrated through a defined API later.

How are locations obtained? The interface uses configured latitude/longitude values. A field system should supply GPS-tagged records and validate coordinates before dispatch.

Are the status changes saved permanently? They currently update client-side state. Server persistence and audit history are next engineering requirements.

How was accuracy measured? No computer-vision accuracy or geolocation-error study has been conducted for this implementation. Avoid claiming accuracy metrics.

How does this relate to Smart Cities? It supports a digitally organized municipal road-maintenance workflow; public value must be evaluated in a future pilot.

Is the interface patentable? No legal novelty claim is made. Public prior art covers road-damage sensing, geolocation, mapping and maintenance reporting; an IP professional must review the detailed claims.

7. Evidence to collect before submission

- Capture dated screenshots of the map, filtered table, detail drawer, KPI views and status transitions.
- Record the browser and screen size used for each test and maintain a pass/fail log.
- Add an API/database test only after those components exist; do not describe planned items as implemented.
- For a future detection evaluation, use a labeled, location-aware holdout set and report precision, recall, F1, latency and coordinate error.
- Insert all team members, institution email, faculty guide, and approved project naming before submission.

Suggested references

[1] H. Maeda, Y. Sekimoto, T. Seto, T. Kashiyama, and H. Omata, “Road damage detection and classification using deep neural networks with smartphone images,” Computer-Aided Civil and Infrastructure Engineering, vol. 33, no. 12, pp. 1127–1141, 2018, doi: 10.1111/mice.12387.

[2] D. Arya et al., “RDD2022: A multi-national image dataset for automatic road damage detection,” Geoscience Data Journal, vol. 11, no. 4, pp. 846–862, 2024, doi: 10.1002/gdj3.260.

[3] S. H. Abdelwahed et al., “Advancements in real-time road damage detection: a comprehensive survey of methodologies and datasets,” Journal of Real-Time Image Processing, vol. 22, Art. no. 137, 2025, doi: 10.1007/s11554-025-01683-1.

[4] Leaflet contributors, “Leaflet Quick Start Guide,” Leaflet documentation. [Online]. Available: https://leafletjs.com/examples/quick-start/ (accessed Oct. 11, 2026).

[5] OpenStreetMap contributors, “Copyright and License.” [Online]. Available: https://www.openstreetmap.org/copyright (accessed Oct. 11, 2026).

[6] United Nations, “Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable.” [Online]. Available: https://sdgs.un.org/goals/goal11 (accessed Oct. 11, 2026).

[7] Ministry of Housing and Urban Affairs, Government of India, “Smart Cities Mission Achievements,” Press Information Bureau, Dec. 18, 2024. [Online]. Available: https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=2085711.